

PROGRAM CONTROL direct mode

RUN Start program at lowest line; **RUN 100** from a line

LIST Show program; **LIST 10,50** range

NEW Erase program from memory

CLEAR Zero all variables, keep code

DEL **DEL 10,40** delete line range

CONT Resume after **STOP / ^C**

STOP Halt w/ "BREAK IN line"

END Halt silently

TRACE / NOTRACE show line #s as run

BASIC STATEMENTS

PRINT ? ; joins, , tabs to next field

INPUT **INPUT "NAME?";N\$**

GET Read 1 keypress, no **RETURN**

LET Assign (optional): **A=5**

REM Comment to end of line

GOTO Jump to line #

GOSUB Call; **RETURN** comes back

POP Discard last **GOSUB** return

FOR **FOR I=1 TO 9 STEP 2 ... NEXT**

IF...THEN **IF A>5 THEN 200**

ON **ON X GOTO/GOSUB a,b,c**

DATA **READ / RESTORE** rewinds

DIM **DIM A(20),N\$(5)** arrays

DEF FN **DEF FN S(X)=X*X**

ONERR **ONERR GOTO n** trap errors

Multi-statement line: separate with : e.g. **A=1: B=2: PRINT A+B**

OPERATORS & FUNCTIONS

+ - * / ^ math; **^** = power

= < > <= >= <> compare (true=1)

AND OR NOT logical

INT ABS SGN whole / magnitude / sign

SQR EXP LOG root, **e^x**, **ln**

SIN COS TAN ATN radians

RND(1) random 0-1; **RND(-n)** seed

LEN(A\$) string length

LEFT\$ RIGHT\$ MID\$ **MID\$(A\$,3,2)**

STR\$ VAL num↔string convert

CHR\$ ASC code↔char

PEEK(a) read byte; **POKE a,v** writes

PDL(n) paddle 0-3 → 0-255

FRE(0) free bytes (call to **GC** strings)

SCRN(x,y) lo-res color at cell

TEXT SCREEN & CURSOR

HOME Clear screen, cursor top-left

TEXT Back to 40-col text mode

HTAB n Cursor column 1-40

VTAB n Cursor row 1-24

TAB(n) SPC(n) in **PRINT** only

INVERSE Black-on-white text

FLASH Flashing text

NORMAL Restore normal text

SPEED=n 0 slow ... 255 fast print

Text window pokes: **POKE 32,L** left · **33,W** width · **34,T** top · **35,B** bottom. Cursor pos **POKE 36,col** / **37,row**.

LO-RES GRAPHICS 40x48, GR

GR Enter lo-res (mixed text)

COLOR=n Set color 0-15

PLOT x,y Dot, x 0-39, y 0-47

HLIN a,b AT y Horizontal line

VLIN a,b AT x Vertical line

0 Black	1 Magenta
2 Dk Blue	3 Purple
4 Dk Green	5 Grey
6 Med Blue	7 Lt Blue
8 Brown	9 Orange
10 Grey2	11 Pink
12 Green	13 Yellow
14 Aqua	15 White

HI-RES GRAPHICS 280x192, HGR

HGR Pg1 hi-res + 4 text lines

HGR2 Pg2 full-screen hi-res

HCOLOR=n Color 0-7 (see below)

HPLLOT x,y x 0-279, y 0-191

HPLLOT TO x,y Line from last point

HPLLOT x,y TO .. Chain segments

DRAW s AT x,y Shape s from table

XDRAW XOR draw (erase/move)

ROT= SCALE= Rotation 0-63 / size

SHLOAD Load shape table (tape)

HCOLOR: 0blk1 1grm 2vio 3wht1 4blk2 5org 6blu 7wht2 · **POKE 230,32**=pg1, **64**=pg2.

DOS 3.3 COMMANDS

CATALOG List disk files

LOAD / SAVE BASIC program to/from disk

RUN name Load & run program file

DELETE Remove a file

RENAME a,b Rename file

LOCK / UNLOCK Write-protect a file

VERIFY Check file integrity

INIT HELLO Format disk + greeting prog

BLOAD/BSAVE Binary; **A\$addr,L,\$len**

BRUN Load & exec binary

OPEN/CLOSE Text file access

READ/WRITE to open file

APPEND Add to end of text file

POSITION Skip records (R param)

EXEC Run commands from text file

MON / NOMON Show/hide disk I/O: C,I,O

MAXFILES n Open-file buffers

FP / INT Applesoft / Integer BASIC

PR#s IN#s Route out/in to slot s

From a program: **PRINT CHR\$(4); "CATALOG"** (**^D** = **CHR\$(4)**). Boot a disk: **PR#6**.

PRODOS / BASIC.SYSTEM

CAT / CATALOG Short / full directory

PREFIX /VOL/DIR Set or show current path

-PATHNAME Run any file (BAS/SYS/BIN)

CREATE Make subdirectory

RENAME a,b Rename / move

LOCK/UNLOCK Protect file

FLUSH Write buffers to disk

STORE/RESTORE Save/load variables

CHAIN Run prog, keep variables

BYE Quit to program selector

Paths use **/VOLUME/DIR/FILE**. Most DOS 3.3 file verbs (**LOAD, SAVE, BLOAD...**) also work.

MACHINE MONITOR CALL -151

aaaa Examine byte at \$aaaa

aaaa.bbbb Dump memory range

aaaa:v v v Store bytes from \$aaaa

aaaal Disassemble 20 instrs

aaaaG Go / run code at \$aaaa

d<s.eM Move (copy) block to d

d<s.eV Verify block vs d

^E Show A,X,Y,P,S registers

n+m n-m Hex add / subtract

^Y Call user routine @ \$03F8

Exit: **^C+RET** → BASIC. Useful G-jumps: **3D0G** reconnect DOS · **C600G** boot slot 6 · **E000G** cold Applesoft · **FF69G** monitor.

KEY CHR\$ CODES

CHR\$(4) **^D** — DOS command flag

CHR\$(7) Bell / beep

CHR\$(8) Backspace (cursor left)

CHR\$(13) **RETURN** (carriage return)

CHR\$(32) Space; **CHR\$(34)**=quote "

ASC("A") = 65; digits "0"=48

SOFT SWITCHES & I/O PEEK/POKE

\$C000 / -16384 Keyboard; bit7=1 if key ready

\$C010 / -16368 Clear keyboard strobe

\$C030 / -16336 Toggle speaker (click)

\$C050 / -16304 Graphics mode on

\$C051 / -16303 Text mode on

\$C052 / -16302 Full screen (no text)

\$C053 / -16301 Mixed (4 text lines)

\$C054 / -16300 Display page 1

\$C055 / -16299 Display page 2

\$C056 / -16298 Lo-res

\$C057 / -16297 Hi-res

Read a switch with **X=PEEK(...)**; set with **POKE ...**, 0. Wait for key: **WAIT -16384,128**.

HANDY CALL ROUTINES

CALL -151 Enter machine monitor

CALL -936 HOME — clear text screen

CALL -958 Clear to end of screen

CALL -868 Clear to end of line

CALL -922 Output a carriage return

CALL -198 Ring the bell (beep)

CALL 768 Run code POKEd at \$300

Quick beep without **CALL**: **PRINT CHR\$(7)**.

EDITING & CONTROL KEYS

^C Break running program

^S Pause/resume **LIST** or output

^X Cancel the line being typed

^G Bell

^RESET Reset / break to monitor or BASIC

← → Back/forward copy over text

ESC @ Home & clear screen

ESC E / F Clear to end of line / screen

ESC I J K M Cursor up/left/right/down (II+)

ESC ↑↓←→ Move cursor, no erase (Ile)

After **ESC** the arrows move the cursor without changing text — press a normal key to leave escape mode.

BOOTING & RESET

PR#6 Boot disk in slot 6, drive 1

PR#5 / PR#7 Boot other controller slots

IN#0 PR#0 I/O back to keyboard/screen

#+^RESET Reboot (Ile/Iic, Open-Apple)

#+♦+RESET Self-test diagnostics

No disk? You land in BASIC (I) or the monitor (*). Type **PR#6** to boot.

PROMPTS YOU'LL SEE

I Applesoft BASIC ready

> Integer BASIC ready

***** Machine monitor

? Awaiting **INPUT**

Common errors: **?SYNTAX ERROR**, **?OUT OF DATA**, **?UNDEF'D STATEMENT** (bad line #), **?ILLEGAL QUANTITY** (bad arg), **?OUT OF MEMORY**.

MEMORY MAP 64K

\$0000-FF Zero page (system + ptrs)

\$0100-1FF 6502 stack

\$0200-2FF Keyboard input buffer

\$0300-3FF Free / vectors / your code

\$0400-7FF Text + lo-res page 1

\$0800-BFF Text page 2; BASIC starts \$0801

\$2000-3FFF Hi-res page 1

\$4000-5FFF Hi-res page 2

\$C000-CFFF I/O & slot soft switches

\$D000-F7FF Applesoft / Integer ROM

\$F800-FFFF Monitor ROM